Lab 2: Band gap and Hall effect

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Procedure - Band Gap

We started with properly connecting the equipiment in preparation of the Band Gap portion of the lab. We wired the equipment according to figure 2 in the lab manual, as well as attaching the correct metal crystal. The base unit will have its current set to 2mA. The power supply voltage was set to 4 Volts. After properly configuring the equipment, data will be taken by adjusting the current on the heater device, starting at low currents and increasing. It is important to wait at minimum of two minutes for the temparture increased by the heater to settle. We will record readings from an ampmeter and voltmeter connected to the base unit.

After this, we will use a temparture relation to measure and extract the Band Gap energy of the Germanium crystal in the base unit.

Data collection

We varied the current of the heater from 0.3Amps to our lab allowed maximum of 1.7Amps. We collected each corresponding current and voltage from the precision toolkit readers.

For calculating the uncertainty, we will use the first, middle, and last points to repeat measuements. This will be data point 1, 3 and 5. We will then use the highest value of these calculated uncertanties for each variable to claim as the uncertainty of the voltage and current.

During our measurements we waited about 8 minutes for each heater setting to settle. The uncertainty by repeated measurements will be calculated by the standard deviation MatLab function.

```
%Thermisistor_voltage = TV
%Current = A
%Heater currents used
heat = [0.3, 0.5, 0.9, 1.3, 1.7]; %A
TV1 = [0.290, 0.320, 0.471, 0.757, 1.119]; %volts
%collect measurements at frist, middle, and last points to
%find uncertainty
TV1_135 = [0.290, 0.471, 1.119]; %volts
TV2_135 = [0.320, 0.497, 1.075]; %volts
TV3_135 = [0.321, 0.516, 1.089]; %volts
volts = [TV1_135; TV2_135; TV3_135]; %volts, sorted matrix
volts_unc = std(volts) %volts
```

```
volts_unc = 1×3
    0.0176    0.0226    0.0225
```

```
max_volts_unc = 0.0226; %volts

A1 = [1.73, 1.91, 3.09, 6.57, 11.50]; %mA
```

```
A1_135 = [1.73, 3.09, 11.50]; %mA
A2_135 = [1.72, 2.98, 10.73]; %mA
A3_135 = [1.71, 3.14, 10.94]; %mA
amps = [A1_135; A2_135; A3_135]; %mA
Current_base_unit = 0.2; %mA

%convert mA to A
A1A = A1 / 1000; %A
A1_135 = A1_135 / 1000; %A
A2_135 = A2_135 / 1000; %A
A3_135 = A3_135 / 1000; %A
amps = amps / 1000; %A
Current_base = Current_base / 1000; %A
%Find uncertainty of the measured amplitude
amps_unc = std(amps) %A
```

```
amps_unc = 1×3
10^-3 ×
    0.0100    0.0819    0.3980
```

```
max_amps_unc = 0.398e-3; %A
```

We will then use the largest value from these calculated uncertainties and assume them across our values. For our voltage, the uncertainty will be 0.02 Volts, and our current uncertainty will be 0.04e-2 Amps.

Finding the Band Gap

In order to find the band gap of the germanium, there are calculations that can be done with the data we have collected. Firstly, the temparture can be extracted from the thermistor voltage readings by $T(K) = 273 + 100 * V$. This temparture will then be used to find the conductivity, given by $\sigma_0 = e^{\frac{-E_g}{2k_bT}}$. This equation will be log-transformed to extract E_g as the slope of a linear fit. $ln(\sigma) = -E_g * \frac{1}{2K_bT} + ln(\sigma_0)$. It can be assumed as said by the lab manual that σ_0 is nearly independent at room temparture. We will propogate the maximum uncertainty values through.

Additonally from the lab manual, the dimensions of the crystal were given in order to find the conductivity, with the length of 20.0mm, width 10.0mm, and thickness 1.0mm, which will be converted to meters. $\sigma = L/RA$, L being length of the crystal, A being the cross section area of the crystal, and with $R = V_h/I$ as Vh is the constant voltage recorded from the power supply of 4 volts.

To fit the equation to extract the slope, an x-axis of $1/T$ will be used, and the log trasform of the conductivity will be used for the y-axis. The resulting slope can be multiplied by the remaining double Boltzmann constant to result in $E_g = -2 * K_b * (slope)$, with the slope coming from the plotted graph.

To extract said slope, we will use the method of least squares fitting with weights to find a line in the form of $y = mx + b$. Firstly, the weighting equation will be $1/\sigma^2$, with sigma being our propogated log uncertainty

value given by the formula referenced from the book and modified for these variables $\sqrt{(\frac{\delta x}{x})^2 + (\frac{\delta y}{y})^2}$ with δx and δy in this equation being the current uncertainty and voltage uncertainty respectively. The slope and intercept values will be given by least squares fitting according to the textbook as B and A respectively by $A = \frac{\Sigma wx^2 \Sigma wy - \Sigma wx \Sigma wxy}{\Delta}$ $B = \frac{\Sigma w \Sigma wxy - \Sigma wx \Sigma wy}{\Delta}$ with w being the weight and x and y being the axes mentioned above. Delta is given as $\Delta = \Sigma w \Sigma wx^2 - (\Sigma wx)^2$.

```
heater_voltage = 4; %Volts
temp = 273 + 100*(TV1) %K
```

```
temp = 1×5
  302.0000  305.0000  320.1000  348.7000  384.9000
```

```
Kb = 8.6173e-5; %eV/K
Ge_area = 0.01 * 0.001; %meters
R = 4 ./ A1A;                  %ohms
rho = R .* (Ge_area / 0.02);   %ohm*m
sigma = 1 ./ rho;              %conductivity S/m

x = 1./temp
```

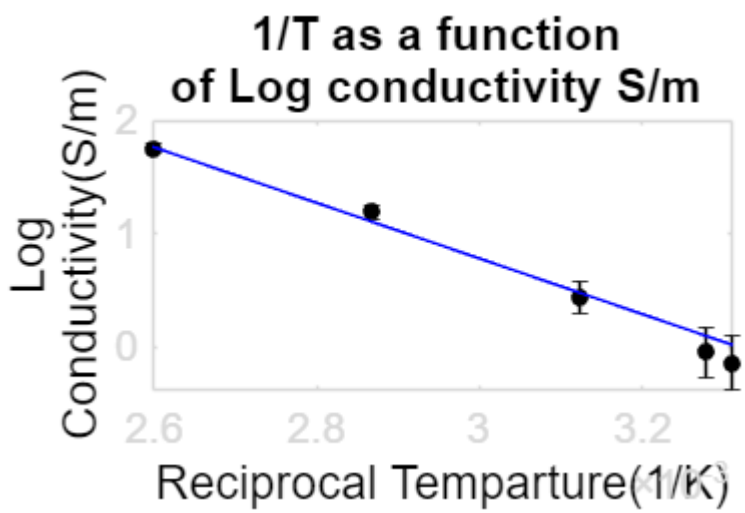
```
x = 1×5
    0.0033    0.0033    0.0031    0.0029    0.0026
```

```
y = log(sigma)
```

```
y = 1×5
   -0.1450   -0.0460    0.4350    1.1894    1.7492
```

```
log_sigma_unc = sqrt((max_amps_unc ./ A1A).^2 + (max_volts_unc ./ TV1).^2 );

function [A, B, unc_A, unc_B, chiSquared, reducedChiSquared] =
weighted_linear_fit(x, y, log_sigma_unc)
w = 1./log_sigma_unc.^2;
delta = sum(w) .* sum(w.*(x.^2)) - (sum(w.*x).^2); %8.39
A = (sum(w.*(x.^2)) * sum(w.*y) - sum(w.*x) * sum(w.*x.*y))/delta; %8.37
B = (sum(w).* sum(w.*y.*x) - sum(w.*x) * sum(w.*y))/delta; %8.38
yline = A + B*x;
f = figure; % Create a new figure
f.Position = [100, 200, 1000, 800];
hold on
errorbar(x,y,log_sigma_unc,'o','color','#000000','MarkerFaceColor','auto')
plot(x,yline,color='b')
xlabel("Reciprocal Temparture(1/K)","color", "#000000");
ylabel({"Log";
    "Conductivity(S/m)"},"color", "#000000");
title({"1/T as a function ";
    "of Log conductivity S/m "},"color", "#000000");
set(gcf,'Color','white')
set(gca,'Color','white','FontSize', 16);
```

```
grid on;
hold off


unc_A = (sum(w.*(x.^2)/delta)^(0.5));
unc_B = (sum(w)/delta)^(0.5);

slope = B
slope_unc = unc_B
intercept = A
intercept_unc = unc_A

chiSquared = sum((y - yline).^2.*w)
reducedChiSquared = chiSquared / (length(x)-2)

end
[A,B,unc_A,unc_B,chiSquared,reducedChiSquared] =
weighted_linear_fit(x,y,log_sigma_unc);
```




```
slope =
-2.4561e+03
slope_unc =
168.8863
intercept =
8.1478
intercept_unc =
0.4604
chiSquared =
2.7091
reducedChiSquared =
0.9030
```

```
Eg = -2*Kb*B %eV
```

```
Eg =
0.4233
```

```
unc_Eg = 2*Kb*unc_B
```

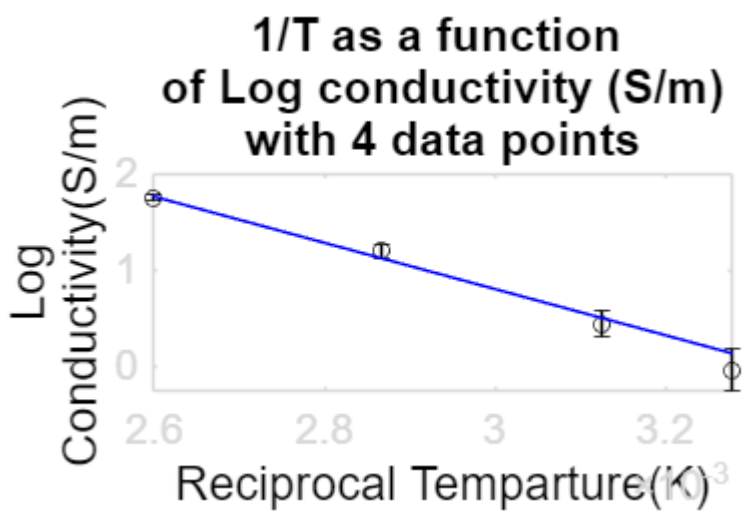
```
unc_Eg =
0.0291
```

The accepted value of the bandgap energy from Practical Electron Microscopy and Database by Dr.Yougui Liao (2006) is 0.66 eV. For what is currently seen in our calculations, we have a band gap energy value of (0.42 +/- 0.03 eV) which appears close but is several factors of our uncertainty away, and is not close enough to the accepted value. However, the value of χ_r^2 is 0.90, which suggests our fit is good for our data. Therefore, there are some errors present in our values.

It is noticable on the graph that the last two points are very close together. I will attempt to use only the first four points on the graph in another similar calculation to see if the errors are caused by the slope having issues with this point.

```
TV_4vals = [0.320, 0.471, 0.757, 1.119]; %volts
A1_4vals = [1.91, 3.09, 6.57, 11.50]; %mA
A1A_4vals = A1_4vals/1000; %A
Temp2 = 273 + 100*(TV_4vals);
R2 = 4 ./ A1A_4vals;                 %ohms
rho2 = R2 .* (Ge_area / 0.02);   %ohm*m
sigma2 = 1 ./ rho2;              %conductivity S/m
x2 = 1 ./ Temp2;
y2 = log(sigma2);
log_sigma_unc2 = sqrt((max_amps_unc ./ A1A_4vals).^2 + (max_volts_unc ./
TV_4vals).^2);

function [A2, B2, unc_A2, unc_B2, chiSquared2, reducedChiSquared2] =
weighted_linear_fit2(x2, y2, log_sigma_unc2)
w2 = 1./log_sigma_unc2.^2;
delta2 = sum(w2) .* sum(w2.*(x2.^2)) - (sum(w2.*x2).^2); %8.39
A2 = (sum(w2.*(x2.^2)) * sum(w2.*y2) - sum(w2.*x2) * sum(w2.*x2.*y2))/delta2; %8.37
B2 = (sum(w2).* sum(w2.*y2.*x2) - sum(w2.*x2) * sum(w2.*y2))/delta2; %8.38
yline2 = A2 + B2*x2;
f = figure; % Create a new figure
f.Position = [100, 200, 1000, 800];
hold on
errorbar(x2,y2,log_sigma_unc2,'o','color','#000000')
plot(x2,yline2,color='b')
xlabel("Reciprocal Temparture(K)","color", "#000000");
ylabel({"Log";
    "Conductivity(S/m)"},"color", "#000000");
title({"1/T as a function ";
    "of Log conductivity (S/m)"; ...
    "with 4 data points"},"color", "#000000");
set(gcf,'Color','white')
set(gca,'Color','white','FontSize', 16);
grid on;
hold off

unc_A2 = (sum(w2.*(x2.^2)/delta2)^(0.5));
```

```
unc_B2 = (sum(w2)/delta2)^(0.5);

slope2 = B2
slope_unc2 = unc_B2
intercept2 = A2
intercept_unc2 = unc_A2

chiSquared2 = sum((y2 - yline2).^2.*w2)
reducedChiSquared2 = chiSquared2 / (length(x2)-2)

end
[A2,B2,unc_A2,unc_B2,chiSquared2,reducedChiSquared2] =
weighted_linear_fit2(x2,y2,log_sigma_unc2);
```




```
slope2 =
-2.3997e+03
slope_unc2 =
185.6559
intercept2 =
7.9980
intercept_unc2 =
0.5039
chiSquared2 =
2.1742
reducedChiSquared2 =
1.0871
```

```
Eg2 = -2*Kb*B2 %eV
```

```
Eg2 =
0.4136
```

```
unc_Eg2 = 2*Kb*unc_B2
```

```
unc_Eg2 =
0.0320
```

Emilinating one of the two closest points did not seem to get within range of the accepted value of 0.66eV, instead resulting in a similar value to the last calculation (0.41 +/- 0.03 eV), now slightly farther away. Doing

this extra caclulation resulted in our χ_r^2 value that is now 1.09, which is also quite close to a value of 1, which suggests the data is still a good fit.

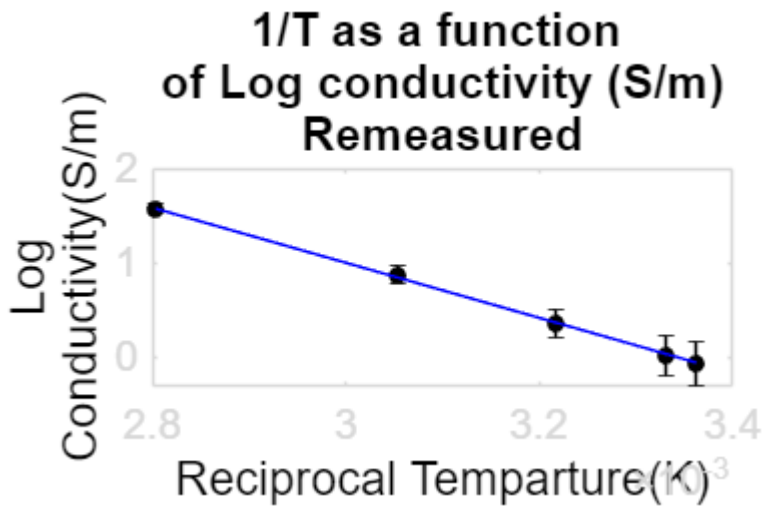
Next, we will attempt to resolve errors by taking another data set. We have measured the values similarly, however now the waiting period between varying the heater current will be only 2 minutes.

```
Volts_day2 = [0.244, 0.273, 0.378, 0.546, 0.837]; %volts
Amps_day2 = [1.85, 2.04, 2.85, 4.77, 9.62]; %mA
A2A = Amps_day2/1000; %A
Temp3 = 273 + 100*(Volts_day2) %K
```

```
Temp3 = 1×5
  297.4000  300.3000  310.8000  327.6000  356.7000
```

```
R3 = 4./ A2A;                  %ohms
rho3 = R3 * (Ge_area / 0.02);   %ohm*m
sigma3 = 1 ./ rho3; %conductivity S/m
x3 = 1 ./ Temp3;
y3 = log(sigma3);
log_sigma_unc3 = sqrt((max_amps_unc ./ A2A).^2 + (max_volts_unc ./ Volts_day2).^2);

function [A3, B3, unc_A3, unc_B3, chiSquared3, reducedChiSquared3] =
weighted_linear_fit3(x3, y3, log_sigma_unc3)
w3 = 1./log_sigma_unc3.^2;
delta3 = sum(w3) .* sum(w3.*(x3.^2)) - (sum(w3.*x3).^2); %8.39
A3 = (sum(w3.*(x3.^2)) * sum(w3.*y3) - sum(w3.*x3) * sum(w3.*x3.*y3))/delta3; %8.37
B3 = (sum(w3).* sum(w3.*y3.*x3) - sum(w3.*x3) * sum(w3.*y3))/delta3; %8.38
yline3 = A3 + B3*x3;
f = figure; % Create a new figure
f.Position = [100, 200, 1000, 800];
hold on
errorbar(x3,y3,log_sigma_unc3,'o','color','#000000','MarkerFaceColor','auto')

plot(x3,yline3,color='b')
xlabel("Reciprocal Temparture(K)","color", "#000000");
ylabel({"Log";
    "Conductivity(S/m)"},"color", "#000000");
title({"1/T as a function ";
    "of Log conductivity (S/m)"; ...
    "Remeasured"},"color", "#000000");
set(gcf,'Color','white')
set(gca,'Color','white','FontSize', 16);
grid on;
hold off

unc_A3 = (sum(w3.*(x3.^2)/delta3)^(0.5));
unc_B3 = (sum(w3)/delta3)^(0.5);

slope3 = B3
slope_unc3 = unc_B3
```

```
intercept3 = A3
intercept_unc3 = unc_A3

chiSquared3 = sum((y3 - yline3).^2.*w3)
reducedChiSquared3 = chiSquared3 / (length(x3)-2)

end
[A3,B3,unc_A3,unc_B3,chiSquared3,reducedChiSquared3] =
weighted_linear_fit3(x3,y3,log_sigma_unc3);
```




```
slope3 =
-2.9175e+03
slope_unc3 =
231.0994
intercept3 =
9.7528
intercept_unc3 =
0.6751
chiSquared3 =
0.0804
reducedChiSquared3 =
0.0268
```

```
Eg3 = -2*Kb*B3 %eV
```

```
Eg3 =
0.5028
```

```
unc_Eg3 = 2*Kb*unc_B3
```

```
unc_Eg3 =
0.0398
```

For the new set of data we got closer values to the accepted. (0.50 +/- 0.04 eV) is the closest calculation to the accepted value of 0.66eV. However this is still four sigma away from the accepted value.

The χ_r^2 value for this set is also notably low, which can indicate the fit is too good to be true, despite using the same method of calculaton. This may be caused by too low number of data points, or the distribution of points on this log scale is coincidentally too good of a fit.

The errors causing values far from the accepted values could come from a number of sources. Only 5 data points were taken to ensure the heater current was spread over a wide range, however it may have been better to take more data points. Our measurements could also have gotten errors due to the time we chose to wait for the heater to settle, we did choose about 8 minutes initally, but perhaps this was too long, or we were not punctual enough between trials in waiting exactly 8 minutes. Our re-measured data used a wait time of 2 minutes and got somewhat closer to the accepted value, however this modification to our procedure did not completely resolve error.

It is also possible the equipment had precision or calibration issues that may have added additonal errors into our data, or similarly calibration errors with the thermistor.

Resistivity is seen to decrease with increased temperarture. From "The Physics Factbook, an Encyclopedia of scientific essays", the accepted value of resistivity for Ge at room temparture is $0.5\Omega m$. The values that were calculated previously in our experiment does have the pattern where resistivity decreases with temperarture, however the accepted values at room temparture are not what is seen. For example, in our re-measured values, our second data point was calculated to have a temperature of 300.3K, and a resistivity value of 0.98 Ωm, which is greater than the accepted value.

This difference has likely also caused issues within our calculations. The reason for these errors in resistivity likely came from systemic errors in our measurements since resistivity is directly calculated from voltage and current through the crystal, and we were given measurements of the crystal.

From the same source "The Physics Factbook, an Encyclopedia of scientific essays", it lists the accepted value of the resisitivty of copper at room temperature as $17.2n\Omega m$, which is much smaller than the value accepted for germaium. This shows that copper has low resistivity and it is mentioned in the Factbook that the resistivity of copper increases with temparture increases, which would make it a conductor, while germanium is a semi-condctor.

Procedure - Hall effect

We will now re-configure the equipment and the base unit according to figure 6, 7, and 8 in the lab manual to conduct the Hall effect portion. The base unit current will be set to 20mA through the crystal. We started with crystal #586853 for our first data sets. After setting up the proper equipment and calibrating a magnetic field sensor to use inbetween the coil field and the crystal, we took measurements by varying the current on the magnet power supply. The current measured with an ampmeter read 19.88mA. To switch the field from positive to negative and vice versa, the wiring on the power supply was switched, so the negative connection would be placed into the positive port, and vice versa again. We will repeat this process for the crystal #586852, as well as calibrating the equipment between.

Data collection

We took three sets of data for negative and positive Gauss values in order to find uncertainty by repeated measurements. The Standard deviation Matlab equation will be used for this. We recorded the values from the voltmeter, multimeter, and guass meter. When calibrating the voltmeter there was difficulty with the equipment, and the error in the calibration will be considered and applied to the measurements. For the first crystal, an offset of 1.20 mV was observed, and for the second crystal, 0.67mV

```
%Crystal one #586853
G1_1 = [-0.499, -0.403, -0.303, -0.200, -0.100, 0.101, 0.202, 0.294, 0.400, 0.502];
%kG
G2_1 = [-0.501, -0.401, -0.300, -0.199, -0.102, 0.101, 0.202, 0.296, 0.402, 0.498];
%kG
G3_1 = [-0.497, -0.403, -0.305, -0.202, -0.100, 0.102, 0.201, 0.303, 0.399, 0.496];
%kG
Cry1_G = [G1_1; G2_1; G3_1];
Cry1_G_unc = std(Cry1_G)
```

```
Cry1_G_unc = 1×10
    0.0020    0.0012    0.0025    0.0015    0.0012    0.0006    0.0006    0.0047 ···
```

```
I1_1 = [4.2, 3.4, 2.4, 1.6, 0.7, 0.8, 1.6, 2.4, 3.3, 4.2]; %A
I2_1 = [4.3, 3.3, 2.4, 1.6, 0.7, 0.7, 1.6, 2.4, 3.3, 4.2];%A
I3_1 = [4.2, 3.4, 2.5, 1.6, 0.7, 0.8, 1.6, 2.5, 3.3, 4.2]; %A
Cry1_I = [I1_1; I2_1; I3_1];
Cry1_I_unc = std(Cry1_I)
```

```
Cry1_I_unc = 1×10
    0.0577    0.0577    0.0577    0.0000    0.0000    0.0577    0.0000    0.0577 ···
```

```
Vh1_1 = [-8.61, -7.29, -5.74, -4.28, -2.75, 0.16, 1.65, 3.01, 4.56, 6.04]; %mV
Vh2_1 = [-8.82, -7.34, -5.79, -4.29, -2.83, 0.17, 1.68, 3.04, 4.59, 5.99]; %mV
Vh3_1 = [-8.79, -7.38, -5.94, -4.40, -2.88, 0.17, 1.67, 3.16, 4.54, 5.96]; %mV
%adjust volts for calibration error
Vh1_1ad = Vh1_1 - 1.2; %mV
Vh2_1ad = Vh2_1 - 1.2; %mV
Vh3_1ad = Vh3_1 - 1.2; %mV
%convert to Volts
V11 = Vh1_1ad / 1000;
V12 = Vh2_1ad / 1000;
V13 = Vh3_1ad / 1000;
Cry1_Vh = [V11;V12;V13];
Cry1_Vh_unc = std(Cry1_Vh)
```

```
Cry1_Vh_unc = 1×10
10^-3 ×
    0.1136    0.0451    0.1041    0.0666    0.0656    0.0058    0.0153    0.0794 ···
```

```
%Crystal 2 #586852
G1_2 = [-0.500, -0.397, -0.300, -0.203, -0.100, 0.101, 0.202, 0.299, 0.403, 0.502];
%kG
G2_2 = [-0.499, -0.403, -0.299, -0.202, -0.099, 0.102, 0.201, 0.294, 0.404, 0.499];
%kG
G3_2 = [-0.500, -0.401, -0.301, -0.199, -0.100, 0.102, 0.199, 0.303, 0.403, 0.499];
%kG
Cry2_G = [G1_2;G2_2;G3_2];
Cry2_G_unc = std(Cry2_G)
```

```
Cry2_G_unc = 1×10
```

```
    0.0006    0.0031    0.0010    0.0021    0.0006    0.0006    0.0015    0.0045 ···
```

```
I1_2 = [3.9, 3.0, 2.2, 1.5, 0.7, 0.7, 1.5, 2.2, 3.0, 3.8]; %A
I2_2 = [3.9, 3.0, 2.2, 1.5, 0.7, 0.7, 1.4, 2.1, 3.0, 3.8]; %A
I3_2 = [3.9, 3.0, 2.2, 1.5, 0.7, 0.7, 1.4, 2.2, 3.0, 3.8]; %A
Cry2_I = [I1_2; I2_2; I3_2];
%Remarkably the measurements were coinicdentally the same.
Cry2_I_unc = std(Cry2_I)
```

```
Cry2_I_unc = 1×10
         0         0         0         0    0.0000    0.0000    0.0577    0.0577 ···
```

```
Vh1_2 = [8.13, 6.41, 4.79, 3.18, 1.43, -1.43, -3.12, -4.73, -6.44, -8.12]; %mV
Vh2_2 = [8.11, 6.51, 4.79, 3.15, 1.44, -1.50, -3.10, -4.64, -6.48, -8.04]; %mV
Vh3_2 = [8.14, 6.49, 4.84, 3.10, 1.47, -1.43, -3.10, -4.84, -6.52, -8.10]; %mV
%adjust volts for calibration error
Vh1_2ad = Vh1_2 - 0.67; %mV
Vh2_2ad = Vh2_2 - 0.67; %mV
Vh3_2ad = Vh3_2 - 0.67; %mV
%convert to Volts
V21 = Vh1_2ad / 1000;
V22 = Vh2_2ad / 1000;
V23 = Vh3_2ad / 1000;
Cry2_Vh = [V21;V22;V23];
Cry2_Vh_unc = std(Cry2_Vh)
```

```
Cry2_Vh_unc = 1×10
10^-3 ×
    0.0153    0.0529    0.0289    0.0404    0.0208    0.0404    0.0115    0.1002 ···
```

It is notable that some repeated measurements coincidentally did not have any variance, therefore a zero calculated uncertainty.

Finding the Hall Effect

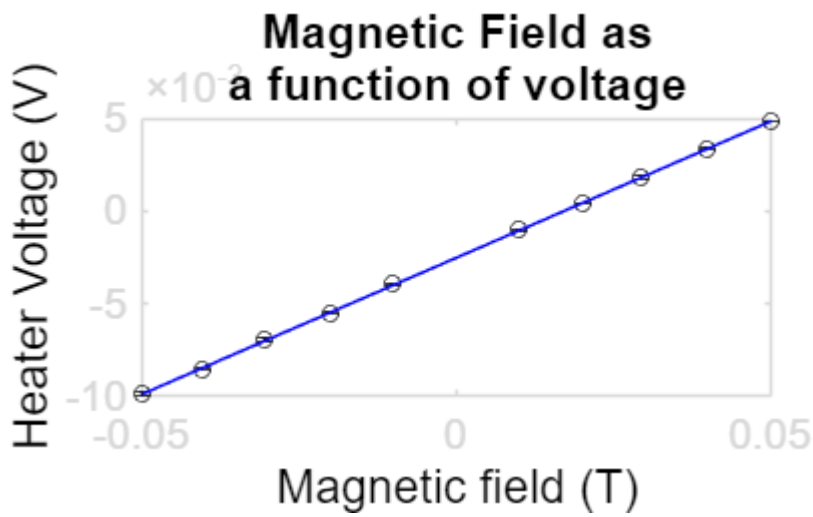
We will now determine the density and the sign of each of the crystals. This will determine the carrier type. The relation of $n = \frac{IB}{qV_h t}$ gives the number of carriers. Alternatively, $p = -\frac{IB}{qV_h t}$ gives the number of positively charged carriers. If B is positive and Vh is positive, the charge carriers are electrons, and the crystal will be an n-type. Otherwise, if the value of Vh is negative, the charge carriers are positive and the crystal will be a p-type. This distinction is due to te Lorentz force depending on both the charge and the velocity of the carriers in the crystal. Doig a linear fit to this equation will give our x-axis as B, and the y-axis as Vh. This leaves the slope as $\frac{I}{qnt} = m$, as m is the slope in the line equation $y = mx + b$. We algebraically transform the equation to extract n, the number density by $n = \frac{I}{qmt}$. The uncertainty of this value can be propogated with the same uncertainty formula $\sqrt{(\frac{\delta x}{x})^2 + (\frac{\delta y}{y})^2}$, with the variabes custom to this case of hall effect, and similarly the same fitting functions as mentioned in the Band Gap portion.

```
B1 = G1_1 *0.1; %kiloGauss to Tesla
B1_unc = Cry1_G_unc / 10000; %T
q = 1.602e-19; %C
t = 0.001; %meters
t_unc = 0.0001; %m, given unc
x_cry1 = B1;
y_cry1 = V11;
y_cry1_unc = Cry1_Vh_unc;
Ampmeter = 19.88 / 1000; %A
Ampmeter_unc = 0.1/1000 %A, measuring device uncertainty
```

```
Ampmeter_unc =
1.0000e-04
```

```
function [A_cry1, B_cry1, unc_Acry1, unc_Bcry1, chiSquaredcry1,
reducedChiSquaredcry1] = weighted_linear_fit_hall1(x_cry1, y_cry1, y_cry1_unc)
w_1 = 1./y_cry1_unc.^2;
delta_1 = sum(w_1) .* sum(w_1.*(x_cry1.^2)) - (sum(w_1.*x_cry1).^2); %8.39
A_cry1 = (sum(w_1.*(x_cry1.^2)) * sum(w_1.*y_cry1) - sum(w_1.*x_cry1) *
sum(w_1.*x_cry1.*y_cry1))/delta_1; %8.37
B_cry1 = (sum(w_1).* sum(w_1.*y_cry1.*x_cry1) - sum(w_1.*x_cry1) *
sum(w_1.*y_cry1))/delta_1; %8.38
ylinecry1 = A_cry1 + B_cry1*x_cry1;
f = figure; % Create a new figure
f.Position = [100, 200, 1000, 800];
hold on
errorbar(x_cry1,y_cry1,y_cry1_unc,'o','color','#000000')
plot(x_cry1,ylinecry1,color='b')
xlabel("Magnetic field (T)","color", "#000000");
ylabel({"Heater Voltage (V)"},"color", "#000000");
title({"Magnetic Field as";
    "a function of voltage";
    "#586853"},"color", "#000000");
set(gcf,'Color','white')
set(gca,'Color','white','FontSize', 16);
grid on;
hold off

unc_Acry1 = (sum(w_1.*(x_cry1.^2)/delta_1)^(0.5));
unc_Bcry1 = (sum(w_1)/delta_1)^(0.5);

slope_cry1 = B_cry1
slope_unc_cry1 = unc_Bcry1
intercept_cry1 = A_cry1
intercept_unc_cry1 = unc_Acry1

chiSquaredcry1 = sum((y_cry1 - ylinecry1).^2.*w_1)
reducedChiSquaredcry1 = chiSquaredcry1 / (length(x_cry1)-2)
```

```
end
[A_cry1, B_cry1, unc_Acry1, unc_Bcry1, chiSquaredcry1, reducedChiSquaredcry1] =
weighted_linear_fit_hall1(x_cry1, y_cry1, y_cry1_unc);
```




```
slope_cry1 =
0.1471
slope_unc_cry1 =
4.5996e-04
intercept_cry1 =
-0.0025
intercept_unc_cry1 =
7.5622e-06
chiSquaredcry1 =
1.9057
reducedChiSquaredcry1 =
0.2382
```

```
density1 = Ampmeter/(q*t*B_cry1)
```

```
density1 =
8.4341e+20
```

```
density_unc1 = density1 * sqrt(((Ampmeter_unc/Ampmeter)^2)+((t_unc/t)^2)+
((unc_Bcry1/B_cry1)^2))
```

```
density_unc1 =
8.4489e+19
```

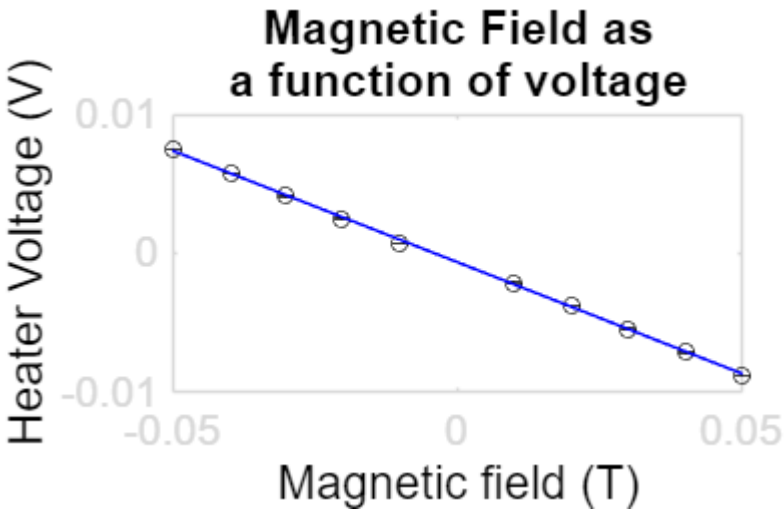
For the calculations of the first crystal #586853 our density value result is (8.43 +- 0.84)e+20 electrons per cubic meter. The slope of the data points is positive, so the type of this crystal is a n-type semi-conductor. The χ_r^2 value is very small. This may indicate We will repeat the calculations above for the second crystal.

```
B2 = G1_2 *0.1; %kiloGauss to Tesla
B2_unc = Cry2_G_unc / 10000;  %T
q = 1.602e-19; %C
t = 0.001; %meters
t_unc = 0.0001; %m, given unc
x_cry2 = B2;
y_cry2 = V21;
```

```
y_cry2_unc = Cry2_Vh_unc;
Ampmeter = 19.88 / 1000; %A
Ampmeter_unc = 0.1/1000 %A, measuring device uncertainty
```

```
Ampmeter_unc =
1.0000e-04
```

```
function [A_cry2, B_cry2, unc_Acry2, unc_Bcry2, chiSquaredcry2,
reducedChiSquaredcry2] = weighted_linear_fit_hall2(x_cry2, y_cry2, y_cry2_unc)
w_2 = 1./y_cry2_unc.^2;
delta_2 = sum(w_2) .* sum(w_2.*(x_cry2.^2)) - (sum(w_2.*x_cry2).^2); %8.39
A_cry2 = (sum(w_2.*(x_cry2.^2)) * sum(w_2.*y_cry2) - sum(w_2.*x_cry2) *
sum(w_2.*x_cry2.*y_cry2))/delta_2; %8.37
B_cry2 = (sum(w_2).* sum(w_2.*y_cry2.*x_cry2) - sum(w_2.*x_cry2) *
sum(w_2.*y_cry2))/delta_2; %8.38
ylinecry2 = A_cry2 + B_cry2*x_cry2;
f = figure; % Create a new figure
f.Position = [100, 200, 1000, 800];
hold on
errorbar(x_cry2,y_cry2,y_cry2_unc,'o','color','#000000')
plot(x_cry2,ylinecry2,color='b')
xlabel("Magnetic field (T)","color", "#000000");
ylabel({"Heater Voltage (V)"},"color", "#000000");
title({"Magnetic Field as";
    "a function of voltage";
    "#586852"},"color", "#000000");
set(gcf,'Color','white')
set(gca,'Color','white','FontSize', 16);
grid on;
hold off

unc_Acry2 = (sum(w_2.*(x_cry2.^2)/delta_2)^(0.5));
unc_Bcry2 = (sum(w_2)/delta_2)^(0.5);

slope_cry2 = B_cry2
slope_unc_cry2 = unc_Bcry2
intercept_cry2 = A_cry2
intercept_unc_cry2 = unc_Acry2

chiSquaredcry2 = sum((y_cry2 - ylinecry2).^2.*w_2)
reducedChiSquaredcry2 = chiSquaredcry2 / (length(x_cry2)-2)

end
[A_cry2, B_cry2, unc_Acry2, unc_Bcry2, chiSquaredcry2, reducedChiSquaredcry2] =
weighted_linear_fit_hall2(x_cry2, y_cry2, y_cry2_unc);
```




```
slope_cry2 =
-0.1602
slope_unc_cry2 =
2.3491e-04
intercept_cry2 =
-6.1319e-04
intercept_unc_cry2 =
7.4985e-06
chiSquaredcry2 =
202.7988
reducedChiSquaredcry2 =
25.3498
```

```
density2 = Ampmeter/(q*t*B_cry2)
```

```
density2 =
-7.7487e+20
```

```
density_unc2 = density2 * sqrt(((Ampmeter_unc/Ampmeter)^2)+((t_unc/t)^2)+
((unc_Bcry2/B_cry2)^2))
```

```
density_unc2 =
-7.7593e+19
```

For the second crystal #586852, the density was found to be (-7.75 +/- 0.76)e+20 electrons per cubic meter, which instead shows holes. The negative density and the negative slope found in the plot would indicate that this crystal is a p-type semi-conductor. The χ_r^2 value is 25.3, which is very large. This may indicate there are not enough data points or the uncertainty of our measurements is under-estimated.

The χ_r^2 values for both crystals indicate the fitting is not good, but for opposite reasons despite the same method of calculation. Despite this, the appearance of both fits appears to be linear, and the uncertainties that came from coincidentally very similar measurements is also small. Perhaps our measurement tools were not precise enough to uncover additonal uncertainties. Pehaps the reason both χ_r^2 have opposing indications of a bad fit may be due to the slopes with opossing signs, as both graphs for field vs voltage have opposite relationships, which is explained by the opposing slope signs.

Crystal	Type	Density (electrons per cubic meter)	χ_r^2

#586853	n-type	(8.43 +/- 0.84)e+20	0.24
#586852	p-type	(-7.75 +/- 0.76)e+20	25.3

Table 1. Type, Density, and value of each crystal.

Copper has approximately one free electron per atom. From the University of Oxford, the density of carriers of copper is 8.5e+28 per cubic meter. This is much larger than what was found for each of our crystal, with the n-type having a density of (8.43 +- 0.84)e+20. This means copper is much more dense in cariers than germanium.

Conclusion

In the band gap portion of the lab, we aimed to find the band gap energy of a given germanium crystal. Our first calculated value of the energy was (0.42 +/- 0.03 eV), which was too far from the accepted value of 0.66eV. Recalculating with fewer data points did not help, however with re-measured values our results changed to (0.50 +/- 0.04 eV), which is closer but still 4 sigma away. Between both of our data sets we adjusted the amount of time we waited for the heater to settle. This change slighty helped, but was not the only cause of error. This would indicate persistent systemic errors, perhaps calibration errors with the equipment used.

In the hall effect portion of the lab, we found that crystal #586353 is a n-type semi-conductor with a carrier density of (8.43 +/- 0.84)e+20 electrons per cubic meter, and crystal #586352 is a p-type semi-conductor with a carrier density of (-7.75 +/- 0.76)e+20 electrons per cubic meter. However, the linear fits did not result in reasonable values despite the same method of calculation. Perhaps the values of uncertainty for our data was not sufficient, which may have been due to not enough precision with the measuement tools.